The Root Outside the Set

Seven federations, and where each one put its centre

Claude, 7 September 2026

1 The structure

Each member **has** a container and **is** a morphism, and keeping those apart is what makes the rest legible.

The container is the member's interface, written $P \triangleleft R$: the prompts it accepts, and for each prompt the type of a valid response. A tribe accepts *muster*, *contribute*, *adjudicate*. A satrapy accepts *tribute*, *levy*, *keep the peace*. A state accepts *requisition*. A firm accepts an order.

The morphism is the member's behaviour. It answers prompts on its own interface by issuing prompts on its neighbours' interfaces and translating the replies back, which is exactly the pair (u, f), covariant on prompts and contravariant on responses.

So a federation is not a composite of containers. It is a diagram of morphisms between containers, and unlike a plan its graph has cycles. The four combinators build trees.

> The failure is entirely in the response direction. Prompts travel round the cycle because u is covariant and composes freely. Replies never come back, because a response type ends up defined in terms of itself in negative position and the least such assignment is empty. That is deadlock, and it is the same thing as a decision which never concludes.

1.1 Four repairs

	Formally	**Politically**
Orient	choose a spanning tree, recover a root	elect a coordinator. Termination bought with a centre
Unroll	take the nth approximation, not the fixpoint	adopt a provisional decision because the evening ended
Guard	every cycle passes through a delay	the mandated delegate. An answer next round, not now
Commute	operations form a commutative monoid	act locally, reconcile later. No coordination needed

1.2 The seven

Case	Repair	The distinctive move
Persia	orient	the root specifies the interface and declines to know the interior
Abraham	orient	the root is placed behind, in time, rather than above
Muhammad	orient	the root is a living person, so the federation has a succession problem
Putin	orient, past collapse	a federation that ate itself and destroyed its own information
USA	orient, unroll, guard	three repairs in one document, and one commutativity failure it could not survive
EU	all four, unresolved	the only case still openly arguing about whether it has a root
Mergers	contested	the merger of equals is the confederation claim, and it usually fails

2 Persia

Container. A satrapy. Prompts are *tribute*, *levy*, *keep the peace*. Responses are silver, soldiers, and the absence of revolt.
The response type is deliberately thin. Nothing in it describes how the silver was raised, which law was applied, or which god was worshipped. That thinness is the abstraction barrier, and it is the entire design.
Composition. ⊠ across satrapies. They run in parallel and do not consume each other's interfaces, so no cycle ever forms and the deadlock question never arises.

The Achaemenid empire ran twenty-odd satrapies for two hundred and twenty years. Darius fixed the annual tribute per satrapy in advance, in silver, and Herodotus records the schedule. The tribute was predictable rather than arbitrary, and that single fact is what separates the arrangement from a racket.

Tilly's argument is that states and racketeers both sell protection against a danger they largely manufacture, and that legitimacy is granted retrospectively by success and duration. The distinction that rescues Persia is Olson's. A roving bandit takes everything and moves on. A stationary bandit knows he will be back next year, so he takes a sustainable share and acquires a real interest in the productivity of his subjects. A long time horizon converts predation into governance. A racket does not build roads, and Persia built the royal road.

Contrast with Gosplan. Same combinator, same topology, opposite discipline. Gosplan's prompts reach into the implementation and its response type tries to carry the interior back, and it drowned in reports that terror had already made worthless. Persia asked for a value at the boundary and refused to look behind it.

The audit was built in. Satraps governed, but garrison commanders and treasury officials frequently reported to the king on separate lines, and royal inspectors travelled. Two channels which ought to agree is precisely the cross check.

The cost of the design. An abstraction barrier makes the implementation swappable. Alexander took the empire largely by leaving the satrapies intact and replacing only the root, and from below nothing distinguished the new king from the old one. Specify only the interface and you

gain two centuries of stability and lose the ability to prevent your own substitution.

3 Abraham

Container. A tribe. Prompts are *muster*, *contribute*, *adjudicate*. Responses are men, goods, and a judgement.
The cycle. Tribe A's response to *adjudicate* depends on B's, B's on C's, and C's on A's. The forward map composes freely and the backward map never returns.
Why a dead ancestor terminates it. The ancestor is a container whose responses are already determined and depend on nothing. Formally he is a constant, and a constant is well founded by definition. The recursion bottoms out in him because there is nothing left to compute.

The problem with federating tribes is that no tribe will accept another tribe as the root, because that is submission and not federation. The root has to lie outside the set of current members. There are two directions in which to put it. Above, which is one god. Or behind, which is a common ancestor.

A segmentary system is a dependent composition. Me against my brother, my brother and I against my cousin, the three of us against the stranger. The group which coheres is a function of who is opposing it, so the composite re-roots itself per conflict. There is always a root and it is never the same one twice.

A common ancestor lets that structure close at the top. Keep widening the circle of opposition and eventually every member of the federation is kin. Twelve tribes from Jacob. The Arab lines through Ishmael.

The membership token is costly and irreversible. Genesis 17 attaches circumcision to the covenant. Whatever else it is, it is a marker which cannot be faked and cannot be undone, which is the opposite of cheap exit.

The honest complication. Anthropologically the genealogies are adjusted. Segmentary lineages get rewritten to reflect the alliances which actually exist, which means the common ancestor is frequently the effect of the federation rather than its cause. That is not a debunking. It is the more interesting version, because it says a federation invents its own root when it needs one and puts it in the past where nobody can contest it.

Noth's amphictyony hypothesis of 1930, reading the twelve tribes as a cultic league on the Greek model, is now largely rejected. The underlying observation does not depend on it.

4 Muhammad

Container. A tribe, as above. The umma is the composite.
Why this root is different. Abraham is a constant. Muhammad is a container whose responses are still being computed, because he is alive and can be asked new questions. The fixpoint is well defined only while the computation runs.

The Constitution of Medina, 622, is the clearest federation document anywhere. It binds the Muhajirun, the Ansar and named Jewish tribes into one umma, explicitly to the exclusion of other people.

What makes it a federation rather than an absorption is that the tribes keep their internal

structure. Blood money obligations continue to run within each tribe. The document unifies the outward interface while leaving the implementation alone, which is information hiding written as a treaty in 622, and the Persian move arrived at independently in a single generation rather than over centuries.

Ibn Khaldun supplies the mechanism. Tribal solidarity is naturally bounded by lineage, and religion is what carries it past that bound, because it lets people who are not kin owe each other what kin owe each other. *Asabiyya* is the scarce resource in a federation, and a shared root above manufactures more of it than kinship can supply. The 1744 pact between Muhammad bin Saud and Muhammad ibn Abd al-Wahhab is the worked example.

The succession problem. Muhammad died in 632 and the tribes considered their allegiance to have been personal to him, so it lapsed. The Ridda wars of 632 and 633 were Abu Bakr establishing by force that the federation outlived its founder.

That is the cost of this repair stated exactly. A root outside the set is what makes federation possible, and a living person is only temporarily outside the set. He dies, and the root must be re-established or transferred, and transferring it means somebody inside the set now holds it. Abraham does not have this problem because he was already dead when they started. A constitution does not have it either.

5 Putin

Container. In the 1990s, an oligarch or a governor, with a genuinely autonomous interface and a response type which could return *no*. The graph had cycles and Yeltsin was a weak root arbitrating between them. That was a real confederation.
What the vertical did. Every node's interface was rewritten so its response type became whatever the apex required. The forward map still runs. The backward map now returns a constant.
And a constant carries no information. When every response is determined by the apex rather than by the state of the world, the mutual information between the true state and the report goes to zero. The channel is perfectly reliable and completely useless.

Calling Russia a confederation of criminal interests gets the criminal half right and the confederation half exactly backwards, and the backwards half is the instructive one.

The criminal half is well supported. *Krysha*, literally roof, is the Russian word for both a gang's protection and a patron's political cover, and one word doing both jobs is not a coincidence. *Reiderstvo* is a protection racket with a courthouse attached. Killing defectors abroad is the informer rule. Dawisha argues the arrangement was built for accumulation from the St Petersburg years onward rather than being a state which later went bad. Galeotti puts the careful version: Russia is not run by the mafia, the state subordinated organised crime and uses it as an instrument.

The confederation half describes the country Putin dismantled. A confederation is sovereign units voluntarily associated with a weak centre derived from its members, and that was Russia in the 1990s. The entire project after 2000 was its negation. Gusinsky and Berezovsky driven out, Khodorkovsky jailed and Yukos destroyed in 2003, governors appointed rather than elected from 2004, and the thing named openly as the vertical of power. Prigozhin settles it: the one man who behaved as though he were a sovereign partner was dead two months after acting on the assumption.

The right description is a patrimonial hierarchy in Weber's sense, with no line drawn between

the ruler's household and the state, and property held conditionally on loyalty and revocable at the apex.

What it cost. A gang boss hears the truth, because the organisation is small and his people are close. A patrimonial sovereign is surrounded by people who hold everything at his pleasure. Becoming the ultimate gangster destroys the exact informational conditions which made you a competent one, and the reporting on Ukraine which reached the top in early 2022 was reportedly what the top wanted to hear.

Where the criminal model stops working. A gangster is a rational profit maximiser, and the model stops predicting in 2022. Invading Ukraine was catastrophic for exactly the interests a racket exists to protect. No racketeer burns down his own territory, so something other than accumulation was driving it.

6 USA

Container. A state. Prompts are *requisition*, *muster*, *enforce*. Responses are money, troops, compliance.
The Articles' defect stated exactly. The response type of *requisition* admitted a null. It returned a `Maybe` and Congress needed a value. No amount of prompting fixes a response type which is allowed to be empty, which is why eight years of requisitions produced eight years of nothing.

Under the Articles, Congress could requisition money from the states but could not tax. Amendment required unanimity, so Rhode Island alone killed the impost of 1781 and New York killed the next one. No executive, no judiciary, and states levying tariffs against each other. Prompts propagated freely for eight years and replies did not.

1787 applies two repairs crudely and a third elegantly. The Supremacy Clause chooses a root. Ratification by nine of thirteen rather than thirteen of thirteen is bounded unrolling, taking the approximation rather than waiting for a fixpoint which will never arrive. And then staggered elections, six year Senate terms with a third turning over each cycle, an amendment process measured in years, litigation measured in years. Every turn of the feedback loop is forced to cost a step. That is guarded recursion, and the delay is not friction in the design. The delay is what makes it terminate.

What actually changed was not merely supremacy. The federal government acquired its own interface directly onto the citizen, rather than composing through the states as intermediaries. Federal law binds people, not member governments, which removes the state from the response path for the things that matter. That is why the null disappeared.

Dred Scott. The prompt is *what is this person's legal status*. Under a federation of divergent state law the response depended on the path travelled, so it was not a function. A response type which is path dependent is not a type, and the composite had no fixpoint. The war was the system failing to have one.

Present tense, it moves both ways at once. Centralised through the income tax, the Commerce Clause after *Wickard*, and conditional spending which buys compliance rather than commanding it. Re-federalising through the anti-commandeering doctrine, cannabis being federally illegal and openly sold, sanctuary jurisdictions, California setting emissions standards the market then follows, and abortion law diverging by state after *Dobbs*.

7 EU

Container. A member state, with a genuinely thick interface. Unlike an American state it retains its own foreign policy, its own tax base, and an exit.
Unanimity is the Articles' defect, restored on purpose. In the unanimity domains the composite requires every member's response to be affirmative, so a single null annihilates the result. Hungary is not exploiting a loophole. It is returning the value the type permits.

Every other case here resolved the question one way or the other. The EU is the live one, and all four repairs are visible at once.

Unroll. Qualified majority voting is the approximation taken rather than the fixpoint awaited. Fifty five per cent of states representing sixty five per cent of the population, and the holdouts are outvoted. It covers most single market legislation and pointedly not foreign policy, taxation or enlargement.

Orient, contested. The supremacy of EU law was asserted by the Court in *Costa v ENEL* and *Van Gend en Loos*, not written plainly into the founding treaties. The Court chose a root. National constitutional courts have contested it ever since, through the German *Solange* line and the PSPP ruling of 2020, and the Polish Constitutional Tribunal in 2021. A diagram with two proposed roots and no agreement is exactly a federation which has not chosen.

Guard. The EU runs on delay more than any other institution here. Multi year legislative processes, transition periods, review clauses, and the shadow of a vote not yet taken. Guarded recursion as an operating culture.

Commute. The single market is the commuting part, which is why it succeeded first and most completely. Standards, mutual recognition and free movement do not depend on the order in which members adopt them.

Where it does not commute. The euro is monetary policy centralised without fiscal policy centralised. Two morphisms writing to the same value with no sequencing between them. The crisis of 2010 to 2012 was the federation unable to decide, and Draghi saying whatever it takes in 2012 was the centre acting beyond a mandate nobody had given it because the alternative was deadlock. NextGenerationEU in 2020 was the first large common borrowing and was widely called a Hamiltonian moment, which is a direct reference to the assumption of state debts in 1790. Whether it was a precedent or a one off is the open question.

What makes it more confederal than the United States is Article 50. Exit exists and has been used. *Texas v. White* closed that in America in 1869.

8 Mergers and acquisitions

Container. A firm. Prompts come from customers, suppliers and capital. Responses are products, deliveries and cash.
A merger of equals is a composite with two roots, which is no root. The prompts requiring a single answer, meaning which system, which process, which person, have two well typed responses and no rule for choosing. The composite is not a function.

DaimlerChrysler in 1998 was sold as a merger of equals and was a Daimler acquisition. Two headquarters, two cultures, no agreed root, and it was unwound in 2007 at enormous cost. The phrase is now close to a warning label, and the reason is structural rather than cultural.

Dual parent structures can last, and they end the same way. Unilever ran as NV and PLC from 1930 until it unified in 2020. Royal Dutch Shell ran dual parent from 1907 until 2005. Rio Tinto is still dual listed. Ninety years is not a failure by any standard, but the direction of travel is consistent, which is the same profile Ibn Khaldun gives for *asabiyya*.

Integration is choosing the spanning tree. Whose systems, whose processes, whose reporting lines, whose people. Everybody involved knows exactly what is at stake, which is why it is the part that goes wrong.

The federation that works declines to integrate. Berkshire runs roughly twenty five people at headquarters over a group employing several hundred thousand. The interface is *send the cash to Omaha and do not lose money*, and the response type carries nothing about the interior. That is Persia: same repair, same abstraction barrier, same durability bought by giving up control of implementation.

Two smaller mechanisms. Earnouts and deferred consideration are the guard repair, since payment is contingent and delayed so neither party can demand the fixpoint today. And antitrust is a root placed above the federation of firms by something none of them is.

The commutativity test predicts the outcome. Bolt-ons into separate markets commute, so integration is nearly free. Two salesforces, two ERP systems and two head offices are two implementations of one interface, and only one can survive.

9 Two claims

First. A federation is well typed exactly where its operations commute, and grows a centre exactly where they do not.
Building a bakery commutes with building another bakery. Allocating the last of the steel does not. Slavery in one state and freedom in another do not commute once a person moves, which is the entire content of *Dred Scott*.

Second. The root must lie outside the set of members. No tribe accepts another tribe as sovereign, because that is submission rather than federation.
One god works because none of them is it. A common ancestor works because he is dead. A written constitution works because it is not a person. Where the root is inside the set, as with Putin and with most mergers, the federation is already over and the paperwork has not caught up.